

Signal-equivalence: testable dose-reduction claims for medical imaging

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Abstract

Claims about reduced-signal medical imaging — “50% dose reduction in CT,” “4× acceleration in MRI,” “25%-activity PET” — are routinely reported as single scalar percentages without specifying the diagnostic task, the patient subpopulation, the equivalence margin, or the statistical confidence over which the claim is meant to hold. The resulting literature is neither comparable across studies nor actionable for clinical decision-making, and the gap is widening as reconstruction methods proliferate.

We introduce **signal-equivalence**, a probabilistic framework that converts every such claim into a 5-tuple credential $(r, T, \varepsilon, \alpha, \Pi)$: a reconstruction method is signal-equivalent to a named reference method at signal-reduction ratio r , task T , equivalence margin ε , and significance level α over patient subpopulation Π if and only if its expected task performance on reduced-signal scans is within ε of the reference method’s performance on full-signal scans, with confidence at least $1 - \alpha$. The framework is modality-general by construction — a single definition covers CT tube-current reduction, MRI k -space undersampling, and PET activity reduction — and inherits its statistical machinery from the test-of-equivalence procedures that the clinical-trial community has used for forty years but that medical-imaging methods papers have not adopted as a reporting standard.

We provide a paired-bootstrap estimator with finite-sample coverage guarantees, a closed-form sample-size formula, and an open-source, pip-installable Python library (`pwm_dose_equivalence`) that produces credentials from any reconstruction method’s image outputs. We demonstrate the framework’s modality-generalizability on seed-reproducible synthetic test sets spanning CT, MRI, and PET using the same single API call, alongside per-modality estimator-coverage simulations; real-cohort credentials for the three worked examples are planned and will populate the per-modality Results tables at Phase 1 and Phase 3a/3b (Section 3). To prevent silent benchmark drift, each credential is content-addressed by the SHA-256 hash of the framework definition it was issued against, so a reader given only a credential JSON can recover the exact framework version and recompute the verdict bit-for-bit.

Adopted as a reporting standard, signal-equivalence converts marketing claims into reviewable scientific propositions that can be verified independently against the published evidence.

1 Introduction

A central question in modern medical imaging is whether a new reconstruction method delivers diagnostically equivalent images at reduced patient exposure or scan time. The question takes a specific form in each modality: in computed tomography it is asked of reduced tube current

(“low-dose CT”); in magnetic resonance imaging of k -space undersampling (“accelerated MRI”); in positron emission tomography of reduced injected radiotracer activity (“low-activity PET”). Across modalities, the dominant evaluation protocol is to report a single scalar metric — peak signal-to-noise ratio (PSNR), structural similarity (SSIM), root-mean-squared error (RMSE) — alongside an unqualified claim such as “50% dose reduction” or “4× acceleration.”

This reporting practice has three interconnected failures.

Failure 1: undefined task. A reconstruction that preserves PSNR may smooth out the diagnostic feature it was supposed to preserve (Kim et al., 2019; Christianson et al., 2015). Without specifying the downstream clinical task — lung-nodule detection, lesion segmentation, organ quantification — the equivalence claim is unanchored. A method may be “dose-equivalent” for lung-nodule detection while being “dose-incompatible” for ground-glass opacity characterization on the same scan.

Failure 2: undefined subpopulation. Methods optimized on adult chest scans frequently degrade on pediatric or oncology-followup populations because the noise statistics, anatomical priors, and lesion-prevalence distributions differ (U.S. Food and Drug Administration, 2024; Larson et al., 2018). An equivalence claim that does not specify the patient distribution over which it holds cannot be transferred to a new clinical context without re-validation.

Failure 3: undefined statistics. “Equivalent” as a binary verdict, without a margin ε and a confidence level α , is not testable. Test-of-equivalence statistics — two one-sided tests (TOST), non-inferiority trial design — have been the standard in pharmacokinetics and clinical-trial design for forty years (Schuirmann, 1987; Piaggio et al., 2012), but have not been adopted as a methodological standard in medical-imaging methods papers.

We argue these three failures share a single solution: every equivalence claim should be a structured, probabilistic, task-conditional, subpopulation-conditional 5-tuple credential. We call this the *signal-equivalence framework*. The framework is intentionally modality-general: rather than producing one definition for CT (with parameter “mA ratio”), a second for MRI (“acceleration factor”), and a third for PET (“activity ratio”), we adopt a single signal-reduction parameter $r \in (0, 1]$ and a modality-specific signal-reduction operator T_r . The framework is then identical across modalities, while each modality’s instantiation of T_r encodes its physics.

Relation to existing equivalence statistics. The statistical machinery is not new: test-of-equivalence procedures such as Schuirmann’s two one-sided tests (TOST) (Schuirmann, 1987) and non-inferiority trial design (Piaggio et al., 2012) have served bioequivalence and clinical-trial endpoint reporting for nearly forty years, and CONSORT (Schulz et al., 2010) requires that every randomized-trial publication report a margin and a confidence level. Medical-imaging *methods* papers have not adopted this discipline. The signal-equivalence framework can be read as TOST applied to a paired-on-patient performance-difference estimator, with the four extra components — explicit task T , explicit subpopulation Π , explicit signal-reduction operator T_r , and a content-addressed credential schema — that are needed for the medical-imaging-methods setting but absent from the pharmacokinetic source material.

Relation to task-based image-quality assessment. A separate, medical-imaging-internal lineage of work has long argued that image quality should be measured against a clinical task, not against scalar fidelity metrics — the canonical statement is Barrett (1990), and the textbook reference is Barrett and Myers (2013), which formalizes the Hotelling and channelized-Hotelling

observer families and the ideal-observer figure-of-merit. AAPM Task Group 233 (Samei et al., 2019) translates this lineage into a CT-specific performance-evaluation standard built around model observers on standardized phantoms. Signal-equivalence is *complementary to*, not a replacement for, task-based image-quality assessment (TB-IQ). TB-IQ scores a single reconstruction-system configuration on a fixed phantom or test set, using a model observer as the figure of merit; it answers “how good is system S ?” Signal-equivalence tests a pairwise equivalence claim between two methods at an explicit signal-reduction ratio over a named patient subpopulation; it answers “does method M at reduced signal match reference method $\mathcal{M}_{\text{ref}}$ at full signal, by how much, and with what confidence?” The two are usefully composed: any TB-IQ figure of merit that produces a real-valued score per case — including the channelized-Hotelling observer of Barrett and Myers (2013) — is a valid choice for $P(T(\cdot), g(\cdot))$ inside a signal-equivalence credential. What signal-equivalence adds on top of the TB-IQ lineage is (a) explicit equivalence-margin and confidence-level reporting, (b) explicit subpopulation as a measure, (c) a signal-reduction operator T_r that ties the claim to a specific physical reduction, and (d) a content-addressed credential format. None of these is present in the TB-IQ standard, and none of the TB-IQ contributions is duplicated here.

We make four separable contributions, each independently testable and adoptable.

1. **A modality-general formalization of equivalence reporting for image-based clinical tasks.** The framework’s mathematical kernel is a paired-on-patient equivalence test; what is novel is the type-signature discipline that ties an equivalence claim to a specific signal-reduction operator T_r defined per modality (Section 4.1), a task with explicit ground-truth protocol, a subpopulation expressed as a measure, and either an aggregate or per-patient formulation. The same definition covers CT tube-current reduction, MRI k -space undersampling, and PET activity reduction — a unification not available in any existing reporting standard or statistics package.
2. **An estimator with finite-sample guarantees and pre-registered sample-size targets** (Section 4.3). The paired bootstrap is conservative for AUC near its boundary; we provide a bias-corrected accelerated (BCa) variant and an exact DeLong test variant for the AUC case, both wired into the same API, with automatic regime detection that warns when the percentile CI is anti-conservative.
3. **An open-source, pip-installable, type-annotated, CI-tested Python library** (`pwm_dose_equivalence`, Apache 2.0) with worked-example notebooks for three modalities (Section 4.4, Section 4.5). Reviewers can install the library, run the notebooks, and reproduce every result in this paper.
4. **A content-addressed credential schema** that ties each published credential to the SHA-256 hash of the exact framework definition it was issued against (Section 4.6). A reader holding only a credential JSON can recover the precise framework version, re-run the same library on the same test set, and verify the verdict bit-for-bit — eliminating a class of benchmark-drift artifacts that has undermined longitudinal comparison in medical imaging (Wang et al., 2020). Content-addressing is the methodological substance of the contribution; the choice of where to host the hash (a local file or a package registry) is a deployment detail discussed in Section 4.6.

A reader can adopt contribution 1 (the framework) without 4 (the registry); contribution 2 (the estimator) without 3 (the library); contribution 3 (the library) without 1 (by writing a different formal definition that the library compiles to). The four contributions are deliberately layered so that each subsequent one strengthens but does not depend on the previous one.

2 Results

Framework overview. We define an *imaging modality* as a triple $\mathcal{M} = (\mathcal{X}, \mathcal{S}, \mathcal{Y})$ consisting of a patient space $\mathcal{X}$, an acquisition space $\mathcal{S}$, and a reconstruction space $\mathcal{Y}$, together with a reference acquisition operator $S_{\text{ref}} : \mathcal{X} \rightarrow \mathcal{P}(\mathcal{S})$ (probabilistic; acquisition is noisy). A *signal-reduction operator at level $r \in (0, 1]$* is a map $T_r : \mathcal{P}(\mathcal{S}) \rightarrow \mathcal{P}(\mathcal{S})$ with the interpretation that $T_r(S_{\text{ref}}(x))$ is the raw measurement distribution one would obtain at reference-level acquisition scaled by r (Section 4.1 formalizes this; the modality-specific instantiations are in Table 6).

The signal-equivalence credential for a reconstruction method M against a reference method $\mathcal{M}_{\text{ref}}$ at the level $(r, T, \varepsilon, \alpha, \Pi)$ asserts that M 's expected task performance on reduced-signal scans is within ε of $\mathcal{M}_{\text{ref}}$'s expected task performance on full-signal scans, with confidence $\geq 1 - \alpha$. The credential is computed via a paired bootstrap over patient draws from Π (Section 4.3); the verdict is one of PASS, FAIL, or INDETERMINATE (CI straddles the equivalence band).

Worked example 1: CT lung-nodule detection at 25% dose. On the public LIDC-IDRI (Armato III et al., 2011) and AAPM-2016 (AAPM, 2016) cohorts and the multi-vendor PWM-LDCT dataset (to be confirmed, 2027a), we will compute signal-equivalence credentials for three baseline reconstruction methods (RED-CNN (Chen et al., 2017), a transformer-based reconstruction, a diffusion-based reconstruction) and the open-source PWM reference method (to be confirmed, 2027b) at $r = 0.25$. Performance metric: lung-nodule detection AUC at fixed FPR = 0.1, with ground truth from majority-vote radiologist annotation. This worked example is scheduled for Phase 1; Table 1 is the placeholder that will carry the credentials (Section 3).

[TODO: TABLE: Signal-equivalence credentials for four CT reconstruction methods at $r = 0.25$, task = lung-nodule detection AUC]

Table 1: Signal-equivalence credentials for four CT reconstruction methods at $r = 0.25$, task = lung-nodule detection AUC (placeholder; populate when data lands).

Worked example 2: accelerated MRI knee-meniscus segmentation at $4\times$. On the fastMRI knee dataset (Zbontar et al., 2020), we will apply the framework with variable-density Cartesian undersampling at $R = 4$ (equivalently $r = 0.25$); the mask family is recorded as part of Π so the 5-tuple shape is preserved (Section 4.1). Performance metric: meniscus segmentation Dice. This worked example is scheduled for Phase 3a; Table 2 is the placeholder that will carry the credentials (Section 3).

[TODO: TABLE: Signal-equivalence credentials for MRI reconstruction methods at $r = 0.25$ ($4\times$ acceleration), task = knee-meniscus segmentation Dice]

Table 2: Signal-equivalence credentials for MRI reconstruction methods at $r = 0.25$ ($4\times$ acceleration), task = knee-meniscus segmentation Dice (placeholder; populate when data lands).

Worked example 3: PET phantom quantification at 25% activity. On the NEMA NU-2 IQ phantom (NEMA, 2018), we will measure contrast-recovery coefficients for the six standard sphere sizes at full and reduced-activity acquisitions. As stated in Data availability, the phantom

acquisition has not yet been performed; this worked example is scheduled for Phase 3b, and Table 3 is the placeholder that will carry the credentials (Section 3).

[TODO: TABLE: Signal-equivalence credentials for PET reconstruction methods at $r = 0.25$ (25% injected activity), task = contrast recovery per sphere size]

Table 3: Signal-equivalence credentials for PET reconstruction methods at $r = 0.25$ (25% injected activity), task = contrast recovery per sphere size (placeholder; populate when data lands).

Cross-modality consistency. The same Python API call signature (Section 4.4) produces a credential for each modality:

```
from pwm_dose_equivalence import signal_equivalence_credential
cred = signal_equivalence_credential(
    method=my_method,
    test_set=load_test_set("AAPM-2016", task="lung_nodule_5mm"),
    reference_method=load_reference("full_dose_FBP"),
    signal_ratio=0.25,
    modality="CT",
    task=Task("lung_nodule_5mm", metric="auc", target=0.95),
    epsilon=0.05, alpha=0.05,
    bootstrap=10_000, seed=42,
)
```

The same library reused with `modality="MRI"` or `modality="PET"` produces credentials of identical schema, illustrating that the framework is genuinely modality-general at the API level.

Cross-modality consistency on synthetic data. The three modality worked examples above are specified with their real-cohort credential tables to be populated at Phase 1 / Phase 3a / Phase 3b (Section 3); at the present manuscript stage they cannot by themselves rule out the possibility that the framework’s modality-generality is rhetorical — three superficially-similar pipelines hand-tuned per modality. To rule that out, we exercise the *same* paired-bootstrap code path on three synthetic test sets, one per modality, each generated under the modality’s T_r from Table 6. For each modality we evaluate two candidate methods: an *equivalent* candidate that is an unbiased estimator of the latent signal with only the additional variance induced by signal reduction, and a *biased* candidate that adds a known additive bias of $0.10 > \varepsilon = 0.05$. With $n = 200$ synthetic patients, $r = 0.25$, $\alpha = 0.05$, and $B = 10,000$ bootstrap replicates at a fixed seed, the same `signal_equivalence_credential` function produces the six credentials in Table 4. All three equivalent candidates pass; all three biased candidates fail; the verdicts are obtained without modality-specific branching anywhere downstream of T_r . The full script and the seed-reproducible JSON output are released at https://github.com/integritynoble/low_dose_CT/tree/main/WS-2_framework/experiments/cross_modality_consistency/.

Comparison with existing tools. Several existing software packages implement components of the proposed pipeline; none implements the full pipeline as a single, modality-general, content-addressed reporting standard. Table 5 summarizes the coverage gap.

Modality	Candidate	$\bar{\Delta}$	CI low	CI high	Verdict
CT	equivalent	-0.0067	-0.0156	0.0022	PASS
CT	biased	0.1011	0.0924	0.1096	FAIL
MRI	equivalent	0.0010	-0.0023	0.0044	PASS
MRI	biased	0.0986	0.0954	0.1018	FAIL
PET	equivalent	-0.0031	-0.0135	0.0076	PASS
PET	biased	0.0915	0.0816	0.1013	FAIL

Table 4: Cross-modality consistency on synthetic data at $r = 0.25$, $\varepsilon = 0.05$, $\alpha = 0.05$, $n = 200$, $B = 10,000$, seed = 42. The same paired-bootstrap function is applied to three modality-specific T_r operators (CT Poisson-thinning, MRI variable-density Cartesian mask, PET list-mode thinning). Equivalent candidates have bias 0; biased candidates have additive bias 0.10. The framework correctly identifies equivalence in all three modalities and correctly rejects in all three; no modality-specific code is exercised downstream of T_r . Numbers are bit-for-bit reproducible from the released script.

Tool	TOST stat	Paired image	Modality-general	Content-a
<code>scipy.stats</code> (Virtanen et al., 2020)	partial	no	no	no
<code>statsmodels</code> (Seabold and Perktold, 2010)	yes	no	no	no
<code>fastMRI evaluator</code> (Zbontar et al., 2020)	no	yes	MRI only	no
<code>MONAI metrics</code> (Cardoso et al., 2022)	no	yes	yes (no T_r)	no
Vendor IR validation reports	no	yes	single-vendor	no
<code>pwm_dose_equivalence</code>	yes (BCa, DeLong)	yes	yes (CT/MRI/PET)	yes (SHA-256)

Table 5: Coverage of the signal-equivalence pipeline by existing tools vs the contribution of this paper. “TOST stat” = test-of-equivalence statistical machinery (BCa / DeLong / percentile bootstrap). “Paired image” = paired-on-patient image-output handling. “Modality-general” = single API call across CT / MRI / PET. “Content-addr.” = content-addressed framework version. “Machine-verifiable” = machine-checkable credential schema.

We built `pwm_dose_equivalence` rather than extending an existing tool for two reasons. First, the type-signature discipline (Methods Section 4.2) requires the framework to know about *both* the statistical estimator and the modality-specific T_r ; no existing tool spans both layers. Second, the content-addressed reporting schema requires a credential format with cryptographic hashing, which falls outside the scope of either statistics or medical-imaging libraries individually.

End-to-end case study: CT lung-nodule detection at 25% dose. To make the workflow concrete, we walk through a single credential computation end-to-end. The user has a candidate reconstruction method `my_method` and wants to know whether it is signal-equivalent to full-dose filtered back-projection on the lung-nodule-detection task at 25% dose.

Step 1 — compute the credential.

```
cred = signal_equivalence_credential(
    method=my_method,
    reference_method=load_reference("full_dose_FBP"),
    test_set=load_test_set("PWM-LDCT", split="test",
                           task="lung_nodule_5mm"),
    signal_ratio=0.25, modality="CT",
    task=Task("lung_nodule_5mm", metric="auc",
```



```

206         ground_truth_protocol=
207             "pwm-ldct/annotation-qa/1.0#sha256:..."),
208         subpopulation="adult_chest_pwm_l3_test_v1",
209         epsilon=0.05, alpha=0.05,
210         formulation="aggregate", estimator="auto",
211         bootstrap=10_000, seed=42,
212     )

```

213 *Step 2 — inspect the resulting credential JSON.*

```

214     {
215         "schema_version": "pwm-signal-equivalence/v0.2",
216         "framework_hash": "sha256:<computed-at-runtime>",
217         "credential": {
218             "method": {"name": "my_method", "code_hash": "sha256:..."},
219             "reference_method": {"name": "FBP_full_dose"},
220             "signal_ratio": 0.25, "modality": "CT",
221             "task": {"name": "lung_nodule_5mm", "metric": "auc",
222                     "ground_truth_protocol":
223                         "pwm-ldct/annotation-qa/1.0#sha256:..."},
224             "subpopulation": "adult_chest_pwm_l3_test_v1",
225             "epsilon": 0.05, "alpha": 0.05,
226             "estimator": "delong", "bootstrap": 10000, "seed": 42,
227             "verdict": "<PASS|FAIL|INDETERMINATE>",
228             "delta_mean": <reported>,
229             "delta_ci_95": [<lower>, <upper>],
230             "n_test": <reported>
231         }
232     }

```

233 *Step 3 — third-party verification.* A reader given only the credential JSON resolves the framework
234 definition document by its SHA-256 hash (from a local file, the pinned PyPI release, or the PWM
235 dataset registry — see Section 4.6), retrieves the test set by its published manifest hash, runs the
236 same library against `my_method`’s published runbundle, and checks that the recomputed verdict
237 and CI match the published values to within Monte-Carlo error of the bootstrap. This is not a
238 hypothetical workflow: it is the acceptance test for every submission to the PWM Low-Dose CT
239 Challenge leaderboard (to be confirmed, 2028).

240 3 Discussion

241 The signal-equivalence framework converts vendor marketing claims into reviewable scientific
242 propositions. A claim of the form “50% dose reduction” is unfalsifiable; a claim of the form “signal-
243 equivalent at ($r = 0.5, T = \text{lung-nodule-AUC}, \varepsilon = 0.05, \alpha = 0.05, \Pi = \text{adult-chest-LIDC}$) against
244 $\mathcal{M}_{\text{ref}} = \text{FBP-at-full-dose}$ ” is falsifiable: re-run the paired-bootstrap on the same Π and the same T ,
245 and the CI either passes or fails the equivalence band.

246 Three implications follow.

247 **Reviewer practice.** A methods paper that does not report a 5-tuple credential, or that reports
248 one with ε or α left unspecified, is not making a comparable claim. Journals adopting the framework

as a reporting standard — analogous to CONSORT for randomized controlled trials (Schulz et al., 2010) — would force authors to make claims that subsequent researchers can verify.

Regulatory practice. The FDA’s draft guidance on AI/ML-based software as a medical device (U.S. Food and Drug Administration, 2024) requires subgroup-stratified performance reporting; a 5-tuple credential per subgroup operationalizes that requirement. A method certified at $(r, T, \varepsilon, \alpha, \Pi_{\text{adult}})$ but not at $\Pi_{\text{pediatric}}$ has explicit evidence of its subpopulation boundary, which is what regulatory reviewers ask for and rarely receive in current submissions. Because each credential is paired with a content-addressed framework hash, it also satisfies the FDA’s emerging predetermined-change-control-plan (PCCP) expectations (U.S. Food and Drug Administration, 2024) for clear delineation of which performance claims are tied to which model version: a credential issued under `framework_hash = sha256:abc...` cannot be silently re-evaluated against a later framework version without producing a new credential under the new hash.

Machine-readable verification. Because a 5-tuple credential is structured and content-addressed, any third party — a human reviewer, a CI script, or an automated evidence-aggregation agent (Anthropic, 2024) — can recompute the verdict from the same test set under the same framework hash and obtain independent corroboration, rather than a re-reading of the abstract. A free-text claim such as “equivalent at 50% dose” supports no such check.

Adoption pathway. Reporting standards in medical imaging typically diffuse in three stages: a methods paper defines the standard; one or two journals require it for a specific subfield (as CONSORT was first adopted journal-by-journal before spreading); and a regulator eventually cites it. Signal-equivalence is positioned for that path — the open-source library and the PWM Low-Dose CT Challenge (to be confirmed, 2028) provide an initial worked-example community, and the subgroup-stratified credential aligns with the FDA’s AI/ML SaMD reporting expectations (U.S. Food and Drug Administration, 2024). Any such trajectory is contingent on the framework’s empirical performance and the library’s maturation, which we will report honestly through the recurring challenge review.

Failure modes of the framework

The framework can return an actionable **PASS** or **FAIL** only when the data and the task are well-specified. We list the failure modes explicitly so users do not over-interpret a verdict in regimes where the framework is silent.

INDETERMINATE verdicts. When the bootstrap CI straddles the equivalence band $(-\varepsilon, \varepsilon)$, the verdict is **INDETERMINATE**: the data are consistent with both equivalence and non-equivalence at the chosen ε and α . We refuse to force a binary verdict in this regime; honest reporting of **INDETERMINATE** is part of the framework, not a defect. **INDETERMINATE** typically indicates one of: (i) insufficient n for the chosen (ε, α) — mitigation: increase n per the sample-size formula; (ii) high per-patient performance variance relative to the margin — mitigation: tighten the subpopulation Π to reduce variance, or relax ε ; (iii) genuinely close-call performance where the method is neither demonstrably equivalent nor demonstrably worse.

Point-evaluated by design. Signal-equivalence is a *point-evaluated* framework: each credential is a separately verifiable claim at a specific $(r, T, \varepsilon, \alpha, \Pi)$, and algebraic combinations of credentials

are deliberately not derivable from one another. This is a design choice, not an omission. Combining credentials algebraically — across signal ratios, tasks, subpopulations, or modalities — introduces unsound chaining risks that have repeatedly produced inflated claims in adjacent fields (Wang et al., 2020); we prefer the conservative position that any new operating point requires its own credential. Four specific claim types are therefore out of scope, by design rather than by oversight:

- *Superiority* (“ M is better than $\mathcal{M}_{\text{ref}}$ at r ”). Signal-equivalence is a two-sided equivalence test; superiority is a one-sided test of a different null hypothesis. Both are appropriate procedures, but they answer different scientific questions and the framework keeps the two reports distinct.
- *Cross-modality transfer* (“ M trained on CT works on MRI”). Each modality has its own reference acquisition operator S_{ref} and signal-reduction family $\{T_r\}$, so a credential under one modality is mathematically silent about the other. Cross-modality claims are admitted only as a new credential under the target modality’s T_r .
- *Composition* (“If $M \equiv \mathcal{M}_{\text{ref}}$ at r and $\mathcal{M}_{\text{ref}} \equiv \mathcal{M}'_{\text{ref}}$ at r' , then $M \equiv \mathcal{M}'_{\text{ref}}$ at rr' ”). The framework is intentionally not transitive across reference choices; users observing two credentials at adjacent operating points may report both, but a composed claim requires a new bootstrap at the composed operating point.
- *Multi-task simultaneity* (“ M is equivalent for tasks T_1 and T_2 ”). A v1 credential is task-specific, which keeps the equivalence margin ε interpretable on a single metric scale; a future task-bundle extension is described in Supplementary [TODO: supp v2].

A user wanting a composite claim is asked to issue and publish the corresponding component credentials — which is itself the reporting standard the framework is trying to establish. The case against a derivable composition law (in particular, why the intuitive $r'' = rr'$, $\varepsilon'' = \varepsilon + \varepsilon'$, $\alpha'' = \alpha + \alpha'$ guess fails) is worked out in [theory/proofs/composition.md].

Estimator failure modes. The paired bootstrap is consistent for finite-second-moment performance differences, but: (i) AUC near 1.0 with small n can produce mildly non-nominal percentile-bootstrap coverage — the library defaults to the closed-form DeLong CI for AUC tasks (mildly conservative; $\sim 300\times$ faster than the bootstrap variants per the coverage simulation in [experiments/estimator_coverage/]) and reports a Monte-Carlo standard error in all cases so users can detect borderline regimes; (ii) heavy-tailed metric distributions can require larger B than the default 10,000 — the same Monte-Carlo SE diagnostic surfaces this; (iii) ground-truth label noise that exceeds ε makes any equivalence verdict meaningless — the library refuses to issue a credential when the documented adjudication-protocol disagreement rate exceeds $\varepsilon/2$.

Subpopulation specificity. A credential under subpopulation Π is a claim about Π specifically and carries no algebraic implication for $\Pi' \neq \Pi$. This is the same point-evaluated discipline as above: where users would like a cross-subpopulation guarantee, the framework asks them to compute and report a separate per-subpopulation credential. The library accordingly does not produce “aggregate” credentials across subpopulations, and the FDA’s subgroup-stratified reporting expectation (U.S. Food and Drug Administration, 2024) aligns naturally with this convention.

Aggregate-vs-per-patient. The aggregate formulation (Definition 1) matches clinical-trial endpoint reporting but is silent about per-patient regret. For settings where the cost of a single missed lesion dominates (rare disease, oncology screening, pediatric), the per-patient strengthening

(Definition 2) should be opted into. The library exposes both formulations through a single `formulation=` argument; the default is `aggregate`.

Reproducibility caveats. A published credential is reproducible only if the test set, the method, the reference method, the bootstrap seed, and the framework version are all reproducible. Content-addressing (Section 4.6) covers the framework definition by its SHA-256 hash, and the test-set manifest by a separate dataset-spec hash; the method’s `code_hash` field covers the method. The library refuses to compute a credential when the method or reference method has not been wrapped in a hash-addressable artifact (Docker image with SHA-256 digest, or pinned PyPI release).

Limitations of the present work

The framework above is methodologically complete — in the sense that the credential schema, the estimator, the sample-size formulas, and the audit surface together form a self-contained reporting standard — but the present manuscript ships with a deliberately scoped empirical-validation surface. We list the limitations explicitly so a reviewer and a downstream adopter can calibrate what we are and are not claiming at this stage. Each limitation paragraph below has a corresponding entry in `paper_draft/limitations_anchors.md` (companion to `paper_draft/reproduction_guide.md`) mapping the limitation to the proofs / simulation document that backs it and to the roadmap milestone that would close it.

Empirical validation is synthetic-anchored, not yet real-cohort. The current evidence backing the modality-general claim is the synthetic cross-modality consistency table (Results, Table 4; 6 credentials at seed = 42 covering CT, MRI, and PET) and the per-modality estimator-coverage simulations (Methods Section ??; 69 coverage cells across AUC, Dice, and contrast-recovery — 27 / 18 / 24 respectively — plus a 24-cell power simulation, 93 simulated (n, σ, ρ) cells in total). The per-modality Results tables (Table 1, Table 2, Table 3) carry **[TODO:]** placeholders that fill at Phase 1 (CT, target D9 + 90 from the PWM-LDCT 1.0 cohort), Phase 3a (MRI, target D9 + 270 via fastMRI), and Phase 3b (PET, target D9 + 270 via the NEMA NU-2 IQ phantom). A reviewer reading the present manuscript should treat the cross-modality consistency demonstration as evidence that the same code path produces sensible verdicts across modalities, not as evidence that the framework correctly discriminates a real candidate method on a real cohort. The reproduction guide (`paper_draft/reproduction_guide.md`) is explicit about which numerical claims are synthetic-anchored versus real-cohort.

Three modalities are validated; the extension surface is not. Methods Table ?? specifies the canonical T_r operator for CT (Poisson thinning of projection counts), MRI (variable-density Cartesian mask, with `mask_family` carried inside Π), and PET (activity reduction); additional modalities — Optical / fluorescence, OCT, ultrasound — are listed as “user-implementable, not validated.” The R3-3 tutorial notebook (`notebooks/04_optical_extending.py`) demonstrates that the API accepts an arbitrary modality string, but no independent research group has yet computed a credential under a user-implemented modality, and the modality-specific operator validity for those extensions remains the user’s responsibility. We expect external adoption (Months 0–12 of the adoption pathway) to surface the first independent extensions; we do not claim the framework is correct for arbitrary modalities at the present manuscript stage.

The credential schema does not yet pin the method bundle. The v0.2 schema carries `method` and `reference_method` as bare-string slugs; the manuscript’s ‘code_hash’ discipline (Sec-

tion 4.1) currently lives *at the publication venue* (a Docker image digest in the supplementary materials, a pinned PyPI release hash, etc.) rather than inside the credential JSON itself. The companion credential reading guide (`paper_draft/credential_reading_guide.md`) and the library’s CONTRIBUTING.md document this as the planned widening at v1.0 (Tier-A schema bump to `{name, code_hash}`); until then, method-bundle provenance is recorded out-of-band and is not enforced by the audit surface. A reviewer evaluating a credential should ask the authors for the matching method-bundle hash and confirm it independently.

Reader-variability is a hard ceiling on attainable ε . The framework’s verdicts are conditional on ground-truth label quality; in the AUC / Dice regime the label-noise floor is whatever the reader-panel adjudication protocol leaves on the table. Concretely: a credential at $\varepsilon = 0.02$ on a task whose inter-reader Dice variance has standard deviation ~ 0.05 is operating below the label-noise floor and the verdict is largely a function of which adjudication subsample fell into the test split, not of the candidate method’s actual signal-equivalence. The library refuses to issue a credential when the documented adjudication-protocol disagreement rate exceeds $\varepsilon/2$ (Section 3, “Estimator failure modes”); however, on tasks where reader-panel disagreement has not been independently quantified, the user is responsible for choosing ε above the plausible label-noise floor. We do not currently provide a per-task label-noise registry; building one is on the v1.0 roadmap.

The headline WS-1 1.0 cohort sits in the INDETERMINATE-dominated regime for the recommended AUC default. At the recommended AUC-task default $\varepsilon = 0.05$ and a typical lung-nodule operating point $\text{AUC} \approx 0.92$, the 1.0 cohort of 208 unique paired patients satisfies the (S3) sample-size formula ($n \approx 130\text{--}250$ required) but the empirical verdict distribution is dominated by INDETERMINATE rather than PASS: $P(\text{PASS})$ under the null is approximately 0.40 at $n = 200$, lifting to approximately 0.94 at $n = 500$ (Section ??, “Cohort-sizing implication”). Downstream users issuing credentials on this cohort should expect INDETERMINATE verdicts for many genuinely equivalent methods; absence of PASS on the 1.0 cohort is not evidence of non-equivalence, only of insufficient n at this (ε, α) . For Dice / MAE / contrast-recovery tasks at the recommended non-AUC default $\varepsilon = 0.02$, the same cohort is comfortably in the PASS regime per the per-modality estimator-coverage simulations. The prospective companion cohort target ($n \geq 500$) lifts the AUC operating point into the reliable-PASS regime. This is a property of the dataset $\times$ operating-point combination, not of the framework itself; we surface it as a limitation so that readers do not misread INDETERMINATE verdicts on the 1.0 cohort as negative results.

4 Methods

4.1 The framework, formally

An imaging modality is a triple $\mathcal{M} = (\mathcal{X}, \mathcal{S}, \mathcal{Y})$ of measurable spaces (patient space, acquisition space, reconstruction space) together with:

- A *reference acquisition operator* $S_{\text{ref}} : \mathcal{X} \rightarrow \mathcal{P}(\mathcal{S})$. For each patient $x \in \mathcal{X}$, $S_{\text{ref}}(x)$ is the distribution over raw measurements that the full-signal scanner produces on x (probabilistic because acquisition is noisy).
- A family of *signal-reduction operators* $\{T_r\}_{r \in (0,1]}$ with $T_r : \mathcal{P}(\mathcal{S}) \rightarrow \mathcal{P}(\mathcal{S})$. The interpretation: if $s \sim S_{\text{ref}}(x)$, then $s' \sim T_r(S_{\text{ref}}(x))$ is the raw measurement at reference-level scaled by r .

Modality	T_r instantiation
CT	Poisson-thinning: if $s_{\text{ref}} \sim \text{Poisson}(\lambda(x))$, then $s_{\text{red}} \sim \text{Poisson}(r \cdot \lambda(x))$
MRI	k -space subsampling: $s_{\text{red}} = \Omega_r \cdot s_{\text{ref}}$ with $ \Omega_r / \Omega_{\text{full}} = r$
PET	Poisson-thinning of list-mode counts at rate r ¹
Optical	Reduced exposure time: $s_{\text{red}} \sim \text{Poisson}(\phi(x) \cdot r \cdot t_{\text{ref}})$

Table 6: Modality-specific signal-reduction operators. CT and PET share the Poisson-thinning model; MRI’s T_r is itself stochastic via the mask distribution.

412 The modality-specific instantiations are given in Table 6.

413 **What “modality-general” means, precisely.** The cross-modality claim has both a mathemati-
414 cal and an empirical component, and we are careful to distinguish them.

415 *Mathematically*, the credential definition (Definition 1) is parametric in $(\mathcal{M}, T_r, T, P)$. Any
416 imaging modality with a well-defined reference acquisition operator S_{ref} and a measurable family of
417 signal-reduction operators $\{T_r\}_{r \in (0,1]}$ admits a credential under this definition. The paired-bootstrap
418 estimator (Algorithm 1) is similarly parametric: it requires only that $P(T(\cdot), g(\cdot))$ produce a real-
419 valued score per case. Both pieces are modality-agnostic by construction; no modality-specific code
420 is required to compile a new modality.

421 *Empirically*, the three modalities we ship as validated have been checked against published
422 baselines and clinically meaningful tasks. Other modalities admit credentials of the same form but
423 have not been validated by us:

Modality	T_r status	Validation status
CT (tube-current reduction)	validated	Phase 1 pilot + multi-vendor PWM-LDCT cohort
MRI (k-space undersampling)	validated	fastMRI knee, variable-density Cartesian mask
PET (activity reduction)	validated	NEMA NU-2 IQ phantom, multiple activity levels
Optical / fluorescence	specified	not validated; user-implementable
OCT (A-scan undersampling)	specified	not validated; user-implementable
Ultrasound (channel sparsity)	not specified	open research question
Diffusion-weighted MRI	special case	requires per-direction T_r design
Spectral / dual-energy CT	special case	requires per-energy T_r design

Table 7: Status of each modality under the framework. “Validated” = a worked example is specified for the modality; the real-cohort credential tables for the three validated examples are populated at Phase 1 / Phase 3a / Phase 3b (Section 3), and the empirical evidence shipped with this manuscript is the synthetic cross-modality consistency demonstration and the estimator-coverage simulations (Section 2). “Specified” = T_r is given in Table 6 but no empirical case-study is included. “Special case” = T_r requires per-axis design beyond the scalar r .

¹The canonical PET T_r for v1 of the framework is *activity reduction* (fewer injected counts at the same scan duration). *Scan-time reduction* at full activity is statistically equivalent at the level of total counts (Poisson-thinning gives the same marginal) but is not physically equivalent — motion blur and kinetic modelling differ — and is documented as a v2 distinction. The synthetic PET channel in [experiments/cross_modality_consistency/] and the worked-example NEMA NU-2 IQ phantom acquisitions both follow the activity-reduction model. See [theory/proofs/pet_reduction.md] for the decision rationale.

What would falsify the modality-general claim. If the framework’s verdicts on MRI or PET disagreed systematically with established clinical-equivalence assessments at the same operating point, the modality-general claim would fail. In the empirical evidence available at this manuscript stage — the synthetic cross-modality consistency demonstration (Section 2) — we do not observe such disagreement; the real-cohort worked examples will provide the corresponding real-data check once their credential tables are populated (Section 3). We call out the falsifiability explicitly because reviewer scrutiny of cross-modality claims is appropriate: any single-modality TOST is uncontroversial, but uniform applicability across physical-acquisition models is the contribution that needs evidence.

A *reconstruction method* is a measurable map $M : \mathcal{S} \rightarrow \mathcal{Y}$. The composed map $M \circ T_r \circ S_{\text{ref}} : \mathcal{X} \rightarrow \mathcal{P}(\mathcal{Y})$ is “reconstruct from a reduced-signal scan of patient x .” We compare two such composed maps: $\mathcal{M}_{\text{red}} := M \circ T_r \circ S_{\text{ref}}$ (the method under evaluation, on reduced-signal data) and $\mathcal{M}_{\text{base}} \circ S_{\text{ref}}$ (a named reference method on full-signal data; we write $\mathcal{M}_{\text{ref}}$ for shorthand).

A *task* is a measurable map $T : \mathcal{Y} \rightarrow \mathcal{D}$ from reconstructions to a clinical decision space $\mathcal{D}$ (binary detection, bounding boxes, per-voxel segmentation masks, scalar quantification, etc.). Ground truth $g(x) \in \mathcal{D}$ is established by a documented adjudication protocol — a board-certified reader panel with a calibration gate and consolidation rule for image-based tasks, or a physical reference standard for phantom tasks — and the protocol identifier is part of the credential. The operational details, including the specific reader-panel discipline used for the CT worked example, are given in Section 4.3.

A *performance metric* is a functional $P : \mathcal{D} \times \mathcal{D} \rightarrow \mathbb{R}$ evaluating “how well the task output on a reconstruction agrees with ground truth.”

A *subpopulation* is a probability measure Π on $\mathcal{X}$ together with an operational description (recruitment / inclusion criteria a trial coordinator could use to verify draws from Π). Crucially, Π also carries the *acquisition-protocol metadata* that fully determines T_r for a member of $\mathcal{M}$ — for CT, the scanner vendor, kVp, reconstruction kernel, and any vendor-specific dose-modulation profile; for MRI, the k -space mask family (uniform random, variable-density Cartesian, equispaced + central-fully-sampled) and the coil sensitivities; for PET, the tracer, scanner, and whether reduction is by injected activity or scan-time. Two subpopulations with the same demographic description but different acquisition protocols are different Π and yield distinct credentials. This convention is what allows the 5-tuple to stay 5-tuple across modalities: modality-specific operational parameters that change the meaning of “signal reduction” (most notably the MRI mask family) are recorded as fields of Π , not as a new tuple slot. The MRI-mask-family case is worked out separately in [theory/proofs/mri_mask.md].

Definition 1 (Signal-equivalence, aggregate formulation). *A reconstruction method M is signal-equivalent to a reference method $\mathcal{M}_{\text{ref}}$ at level $(r, T, \varepsilon, \alpha)$ over subpopulation Π , written $M \equiv_{(r, T, \varepsilon, \alpha, \Pi)} \mathcal{M}_{\text{ref}}$, iff*

$$\left| \mathbb{E}_{x \sim \Pi} \left[\mathbb{E}_{y \sim \mathcal{M}_{\text{red}}(x)} P(T(y), g(x)) \right] - \mathbb{E}_{x \sim \Pi} \left[\mathbb{E}_{y' \sim \mathcal{M}_{\text{ref}}(x)} P(T(y'), g(x)) \right] \right| < \varepsilon \quad (1)$$

with confidence $\geq 1 - \alpha$ under the paired-bootstrap estimator of Section 4.3.

Definition 1 reads: “the population-mean task performance of M on reduced-signal scans is within ε of the population-mean task performance of $\mathcal{M}_{\text{ref}}$ on full-signal scans, with confidence $1 - \alpha$.”

Population claim vs. sample evidence. The two expectations in Equation (1) are deterministic population objects that the credential *asserts*; the role of α and the paired-bootstrap procedure of Section 4.3 is to deliver finite-sample *evidence* for or against that assertion from a test set $\{x_k\}_{k=1}^n$

drawn from Π . The credential schema therefore records, alongside $(r, T, \varepsilon, \alpha, \Pi)$, the test-set size n and a slug resolving to the operational sampling protocol by which draws from Π are produced (e.g. the dataset’s published recruitment criteria and split manifest), so that an independent verifier is guaranteed to be testing the same population claim — not merely recomputing the same bootstrap CI on a possibly-different sample.

We also define a strictly-stronger *per-patient* formulation:

Definition 2 (Signal-equivalence, per-patient formulation). *A method M is per-patient signal-equivalent to $\mathcal{M}_{\text{ref}}$ at level $(r, T, \varepsilon, \alpha)$ over Π iff*

$$\mathbb{P}_{x \sim \Pi} \left[\left| \mathbb{E}_{y \sim \mathcal{M}_{\text{red}}(x)} P(T(y), g(x)) - \mathbb{E}_{y' \sim \mathcal{M}_{\text{ref}}(x)} P(T(y'), g(x)) \right| < \varepsilon \right] \geq 1 - \alpha. \quad (2)$$

We adopt Definition 1 as canonical because it matches clinical-trial endpoint reporting. Definition 2 is an opt-in strengthening (e.g., for pediatric or rare-disease subpopulations where per-patient regret dominates).

4.2 Type signatures: six clarifications

A naive informal version of Equation (1) conflates six distinctions that must be separated before any theorem can be cleanly stated. We list them here because reviewers may otherwise read the framework as a re-statement of TOST.

(C1) Patient vs image. T_r acts on the *acquisition* of a patient, not on the patient. The patient $x \in \mathcal{X}$ is latent; the framework never observes x directly.

(C2) T_r is an operator, not a scalar multiplier. Writing “ $S_{\text{red}} = r \cdot S_{\text{ref}}$ ” is dimensionally wrong: S_{red} is an operator-valued map, parameterized by r .

(C3) Reference acquisition $\neq$ reference algorithm. “Reference” is ambiguous in the informal version. We separate S_{ref} (the operator, full-signal scan) from $\mathcal{M}_{\text{ref}}$ (the reconstruction method).

(C4) Performance is a function of $(T(y), g(x))$, not (M, S, T) . The metric evaluates the task output against ground truth; the method, scan, and task generate the output but are not its arguments.

(C5) Subpopulation is a measure that carries acquisition-protocol metadata, not just a demographic label. Two subpopulations with the same demographic label but different operational definitions — or the same demographic but different scanner / mask / tracer settings — are different subpopulations. This is what lets modality-specific acquisition parameters (CT vendor, MRI mask family, PET reduction mode) live inside Π rather than as extra tuple slots, preserving a single 5-tuple shape across CT, MRI, and PET.

(C6) Aggregate vs per-patient. Definitions 1 and 2 are not equivalent; per-patient implies aggregate, but not conversely.

4.3 Estimator

Ground-truth protocol. Ground truth $g(x_k)$ is the largest single source of credential variance the framework does not control, so the credential records the adjudication protocol by content-addressed reference rather than by free-text slug. For the CT worked example we adopt the PWM-LDCT 1.0 annotation QA protocol (to be confirmed, 2027a) verbatim, summarized here for the reader’s convenience: (i) every annotator is a board-certified radiologist with ≥ 5 years of thoracic CT experience; (ii) each annotator passes a 20-case calibration set drawn from a held-out LIDC-IDRI subset before contributing release annotations, with a pre-registered Cohen’s $\kappa \geq 0.60$ pass threshold

against the LIDC majority vote; (iii) candidate nodules are matched across readers by bounding-box IoU ≥ 0.3 on the same slice with ± 1 -slice 3D adjacency; (iv) a candidate is included in $g(x_k)$ iff more than half of the qualified readers for the case marked it, with attribute consolidation via element-wise median (`bbox`, `diameter_mm`, ordinal `texture`) and mode (`location`); (v) two-reader cases with non-overlapping boxes, $\geq 1.5 \times$ diameter disagreement, or texture-class disagreement are referred to a third independent radiologist whose label becomes the released ground truth; (vi) drift is monitored by a rotating 5-case re-calibration every ≈ 50 cases, and any reader whose running κ falls below the pass threshold is paused and all of their since-last-passing-window cases are re-annotated. The per-reader raw labels are preserved alongside the consolidated ground truth so users of the framework can substitute alternative consolidation rules (e.g. STAPLE) and re-issue the credential without re-running the reader panel. The MRI worked example reuses the published fastMRI expert meniscus segmentations as ground truth, with the segmentation protocol resolved through the same content-addressed mechanism; the PET worked example does not require a reader panel because the NEMA NU-2 IQ phantom provides physical ground truth (known sphere sizes and activity concentrations). In all three cases the credential's `ground_truth_protocol` field resolves to a normative protocol document by SHA-256 hash (Section 4.6), so a downstream verifier can confirm the protocol bit-for-bit rather than from a slug whose meaning may drift.

Label-noise refusal threshold. The library refuses to issue a credential when the documented inter-reader disagreement rate of the cited protocol exceeds $\varepsilon/2$; below that threshold, equivalence verdicts cannot be meaningfully separated from annotation noise. For the CT protocol above, the published pairwise κ on the calibration set is the operational input to this check.

Given a test set $\{x_1, \dots, x_n\}$ drawn i.i.d. from Π with ground truth $\{g(x_k)\}$ established by the protocol above, and methods $M, \mathcal{M}_{\text{ref}}$, we estimate Equation (1) via a paired bootstrap on patients (Algorithm 1).

Algorithm 1 Paired-bootstrap signal-equivalence credential

Require: Test set $\{x_k\}_{k=1}^n$ from Π ; ground truth $\{g(x_k)\}$; methods $M, \mathcal{M}_{\text{ref}}$; reduction operator T_r ; task T ; metric P ; margin ε ; significance α ; replicates B ; seed.

- 1: **for** $k = 1, \dots, n$ **do**
- 2: $s_{\text{red}} \sim T_r(S_{\text{ref}}(x_k)); \quad a_k \leftarrow P(T(M(s_{\text{red}})), g(x_k))$
- 3: $s_{\text{ref}} \sim S_{\text{ref}}(x_k); \quad b_k \leftarrow P(T(\mathcal{M}_{\text{ref}}(s_{\text{ref}})), g(x_k))$
- 4: **end for**
- 5: **for** $b = 1, \dots, B$ **do**
- 6: Sample n indices $\{i_1, \dots, i_n\}$ with replacement from $\{1, \dots, n\}$
- 7: $\Delta_b \leftarrow \frac{1}{n} \sum_{j=1}^n (a_{i_j} - b_{i_j})$
- 8: **end for**
- 9: $[\ell, u] \leftarrow$ central $(1 - \alpha)$ percentile interval of $\{\Delta_b\}$
- 10: **return** verdict $\leftarrow$ **if** $[\ell, u] \subset (-\varepsilon, \varepsilon)$ **then** PASS; **else if** $[\ell, u] \cap (-\varepsilon, \varepsilon) = \emptyset$ **then** FAIL; **else** INDETERMINATE

We use $B = 10,000$ as the default. Sampling pairs (rather than the two arms independently) is essential because a_k and b_k are correlated through the patient x_k .

Sample-size formula. For a target (ε, α) and a per-patient performance-difference standard deviation $\sigma_\Delta = \sqrt{\text{Var}(\Delta_k)}$, the leading-order normal-approximation bound for a sample-mean

estimator is

$$n \geq \left(\frac{z_{1-\alpha/2}}{\varepsilon} \right)^2 \sigma_{\Delta}^2. \quad (3)$$

With per-arm SD σ and cross-arm correlation ρ , $\sigma_{\Delta}^2 = 2\sigma^2(1 - \rho)$, so pairing reduces the required n by a factor $(1 - \rho)/2^{-1}$ relative to two independent arms. For paired AUC, $\hat{\Delta}$ is not a sample-mean estimator; the appropriate specialisation uses the DeLong placement-difference standard deviation s ,

$$n_{\text{AUC}} \geq \left(\frac{z_{1-\alpha/2}}{\varepsilon} \right)^2 \cdot 4s^2. \quad (4)$$

At $\rho = 0.5$ on a binormal model, empirical s values are approximately $s \approx 0.20, 0.15, 0.08$ at AUC $\approx 0.85, 0.92, 0.97$ respectively (full table and derivation in `[theory/proofs/sample_size.md]`). Substituting in (4) gives the ****AUC-task default** ($\varepsilon = 0.05, \alpha = 0.05$), which yields $n \approx 130\text{--}250$ across the typical operating range — comfortably within the WS-1 PWM-LDCT cohort. For Dice, MAE, and per-patient contrast-recovery (where $\hat{\Delta}$ is a sample-mean estimator and (3) applies directly), the canonical default is $\varepsilon = 0.02$; users with a clinical argument for a tighter or looser margin can specify it, and the library will require the corresponding n . A finite-sample Bernstein correction (strictly more conservative than (3)) is documented in Supplementary Section S1 and in `[theory/proofs/sample_size.md]` §4.

Estimator defaults. The percentile bootstrap is the library’s default for general metrics. For paired AUC, the closed-form DeLong test (DeLong, DeLong & Clarke-Pearson 1988) is auto-selected: in 27-cell coverage simulations under the truly-equivalent null ($\Delta = 0$) at AUC $\in \{0.85, 0.92, 0.97\}$ and $n \in \{100, 200, 500\}$, DeLong is mildly conservative (median coverage ≈ 0.96) and approximately 300× faster than the bootstrap variants — a strict improvement over percentile in the AUC case. The bias-corrected accelerated (BCa) variant remains available as an opt-in; under the null it does not robustly outperform percentile (in the same 27-cell sweep BCa coverage falls below the nominal 0.95 at AUC ≥ 0.92 , $n = 100$), so we do not default to it. Full coverage table and the estimator-default justification are in `[theory/proofs/estimator.md]`, backed by the seed-reproducible simulation in `[experiments/estimator_coverage/]`.

4.4 Open-source library

The `pwm_dose_equivalence` library (Apache 2.0; source at https://github.com/integritynoble/low_dose_CT/tree/main/WS-2_framework/pwm_dose_equivalence/, to be published to PyPI at acceptance) implements Definition 1 for CT, MRI, and PET. The user-facing API is a single function call:

```
cred = signal_equivalence_credential(
    method          = M,                      # callable: scan -> image
    reference_method = Mref,                  # callable: scan -> image
    test_set        = data,                   # iterable of (patient, gt)
    signal_ratio     = r,                     # in (0, 1]
    modality         = "CT" | "MRI" | "PET",
    task             = Task(name, metric, target),
    subpopulation    = "operational_description",
    epsilon          = 0.05,                  # AUC default; 0.02 for non-AUC metrics
    alpha            = 0.05,
    formulation      = "aggregate" | "per_patient",
```

```

575         estimator          = "auto" | "percentile" | "delong" | "bca",
576         bootstrap          = 10_000,
577         seed               = 42,
578     )

```

579 Returns a `SignalEquivalenceCredential` object with the verdict (`PASS` / `FAIL` / `INDETERMINATE`),
 580 the bootstrap CI, the sample-size sanity check, and a JSON serialization conforming to the schema
 581 in Section 4.6.

582 4.5 Software engineering rigor

583 The library is intended to be a long-lived community tool, not a paper-companion artifact that
 584 lapses six months after publication. We invested in three classes of engineering quality consistent
 585 with that intent.

586 **Testing.** The current v0.2.2 library ships with 140 tests (10 end-to-end integration tests landed
 587 in v0.2.0; 35 credential-audit tests landed in v0.2.1; 12 CLI tests landed in v0.2.2) covering: the
 588 percentile, DeLong, and BCa CI estimators under controlled σ and ρ ; the (S1) / (S3) / (S4) sample-
 589 size formulas against their numerical-table values and against the coverage simulation; cross-modality
 590 T_r operator invariants (CT and PET Poisson-thinning ratios, MRI mask central-region invariant,
 591 NaN sentinel for unsampled k -space positions); JSON-credential round-trip and content-addressed
 592 framework-hash stability; verdict logic (`PASS` / `FAIL` / `INDETERMINATE`) at the equivalence-band
 593 boundaries; the v0.2.1 internal-consistency audit (`audit_credential`) on tampered, undersized,
 594 BCa-as-headline, and framework-hash-mismatched credentials; the v0.2.2 `pwm-audit` CLI exit-code
 595 contract (0 on ok, 1 on hard issue, 2 on I/O error) across file / stdin / JSON-output / help /
 596 version paths; and explicit failure-mode behaviour (mode-misuse arguments, sample-size warnings
 597 firing when n is below the formula prescription, `ValueError`-raising input validation). Continuous
 598 integration runs on every commit against Python 3.10, 3.11, and 3.12 on Linux runners (macOS
 599 / Windows runners scheduled at v1.0.0); the current line coverage is **100%** on 443 statements
 600 (measured by `coverage.py`), well above the manuscript’s pre-registered $\geq 90\%$ ship floor.

601 **Type discipline.** The codebase is fully typed (PEP 484 annotations enforced by `mypy -strict`),
 602 with explicit `TypedDict` schemas for credential JSON inputs and outputs. The choice of strict
 603 typing is deliberate: when the library is used as a reporting standard, silently-coerced argument
 604 types are a class of error that produces credentials that look valid but are not. Strict types surface
 605 the error at the API boundary.

606 **Documentation and tutorial notebooks.** The library’s documentation (built from the source
 607 tree with Sphinx; to be hosted publicly at acceptance) provides an API reference (auto-generated
 608 from docstrings), three tutorial notebooks (one per validated modality, plus a fourth showing how to
 609 extend the framework to a user-implemented modality at `notebooks/04_optical_extending.py`),
 610 and a “Credential reading guide” (`paper_draft/credential_reading_guide.md`) for reviewers
 611 and regulators who need to interpret published credentials without writing code themselves. The
 612 reading guide walks the credential JSON field-by-field, covers verdict semantics, and lists eight red
 613 flags that should prompt follow-up questions; its technical companion for code-savvy reviewers is
 614 the reproduction guide (`paper_draft/reproduction_guide.md`), which re-derives every numerical
 615 claim in this manuscript from the published artifacts. The notebooks are tested in CI to ensure
 616 they remain runnable as the API evolves.

Versioning policy. Releases follow semantic versioning. MAJOR bumps break the credential JSON schema; MINOR bumps add modalities, estimators, or task types without breaking existing credentials; PATCH bumps are bug fixes that preserve credential reproducibility. The current version is 0.2.2 (alpha; v0.2.0 added BCa + small- n warning + `bound_M` + 10 integration tests; v0.2.1 added `audit_credential` + the machine-readable JSON Schema; v0.2.2 added the `pwm-audit` console entry point); we target 1.0.0 at journal acceptance with a five-year support commitment.

Release cadence and governance. The library is maintained by the PWM Protocol Foundation with a documented contribution policy at `pwm_dose_equivalence/CONTRIBUTING.md`. The policy defines two sign-off tiers: Tier A (two of three maintainer sign-offs) for changes to the methodologically load-bearing modules — `framework_hash.py` (the `FRAMEWORK_SPEC` string whose SHA-256 anchors every credential), `credential.py` and `credential_schema.py` (the wire format), `estimator.py` (the percentile / DeLong / BCa CI variants and verdict logic), `sample_size.py` (the (S1) / (S3) / (S4) formulas), and `operators.py` (the canonical T_r reductions per modality); and Tier B (one sign-off) for documentation, tests, tutorial notebooks, worked examples, audit-rule additions that do not flip an `ok=True` credential to `ok=False`, and API-additive changes. The policy also documents the schema-mutation discipline (`FRAMEWORK_SPEC` edits trigger a MAJOR version bump and require regenerating the standalone `credential_schema.json` and re-issuing the example credentials), the testing floor ($\geq 90\%$ line coverage, currently shipping at 100%), the security-disclosure window for credential-forgery vulnerabilities, and the contributor-license-agreement intent (Apache 2.0). A `CITATION.cff` (CFF v1.2.0) at the same path declares the software citation and the manuscript as the preferred-citation target.

4.6 Content-addressed credential schema

The credential schema is content-addressed: every published credential carries the SHA-256 hash of the exact framework definition document it was issued against. A published credential identifies the framework version by hash:

```
{
  "schema_version": "pwm-signal-equivalence/v0.2",
  "framework_hash": "sha256:...",
  "credential": { ... }
}
```

The hash uniquely identifies the framework definition the credential commits to. A reader verifies the credential by (i) retrieving the framework definition document by hash, (ii) re-running the paired bootstrap on the same test set, and (iii) checking that the recomputed verdict matches the published verdict. This eliminates a class of benchmark-drift artifacts in which a method’s reported performance is silently re-evaluated against a different framework version after submission (Wang et al., 2020).

Internal-consistency audit. A reader who lacks the test-set scores (for example, a regulator without raw-DICOM access) can still check the credential’s *internal* consistency without re-running the bootstrap. The library exposes a one-call audit function, `pwm_dose_equivalence.audit_credential`, which validates the credential JSON against the machine-readable schema (`CREDENTIAL_JSON_SCHEMA`), confirms the framework hash matches the installed library version, re-derives the verdict from the published $(\Delta_{CI,lo}, \Delta_{CI,hi}, \varepsilon)$ and checks that it matches the stored verdict, inspects the `sample_size_check`

field for the undersized-PASS pattern, and flags BCa-as-headline credentials lacking a percentile companion. The audit returns a `CredentialAudit` dataclass distinguishing hard issues (block ok) from soft warnings (recorded but non-blocking). The same audit is available from the shell as the installed console entry point `pwm-audit` (with file / stdin / `-json` input modes and a 0-on-ok / 1-on-issue / 2-on-error exit-code contract), so reviewers, CI hooks, and submission-checklist scripts can verify a credential without writing any Python. Full bootstrap reproduction remains the job of the reproduction guide; the audit is the cheaper preflight check that catches tampered or schema-mismatched credentials before the bootstrap would.

Hash-resolution backends. Content-addressing is the methodological substance; *where* the hash is resolved is a deployment choice. The library accepts any of three resolvers: (a) a local file containing the framework definition (sufficient for self-contained reproduction during peer review); (b) a package-registry resolver that retrieves the definition by hash from the PyPI release used to issue the credential; or (c) an optional public registry mirror. Resolvers (a) and (b) are sufficient for the methodological guarantees stated above; resolver (c) is an operational convenience for community deployments and is not required for the framework to function.

5 Data availability

The three modality validators in this paper use publicly available datasets, all of which require registration but are free for academic use:

- **CT.** The multi-vendor PWM-LDCT dataset ([to be confirmed, 2027a](#)) on PhysioNet credentialled access; supplementary CT data from LIDC-IDRI ([Armato III et al., 2011](#)) (NBIA registration) and the AAPM 2016 Low-Dose CT Grand Challenge ([AAPM, 2016](#)) (Mayo Clinic request).
- **MRI.** The fastMRI knee dataset ([Zbontar et al., 2020](#)) (NYU registration), variable-density Cartesian undersampling subset.
- **PET.** NEMA NU-2 IQ phantom acquisitions at four activity levels (full, 50%, 25%, 10%) are *planned* as a companion data descriptor; the acquisition has not yet been performed, and no PET phantom data are released with this paper. The PET worked example in Section 2 is therefore specified but not populated, and no PET credential in this manuscript rests on measured phantom data.

The exact test-split manifest hashes used for every credential in this paper are listed in Supplementary Table **[TODO: supp tab]**; each hash resolves to a bit-identical manifest via the dataset’s PhysioNet record, the supplementary data archive, or the PWM dataset registry (any of the three is sufficient for reproduction).

6 Code availability

The signal-equivalence library, `pwm_dose_equivalence`, is released under Apache 2.0; the source repository is at https://github.com/integritynoble/low_dose_CT/tree/main/WS-2_framework/pwm_dose_equivalence/. A PyPI release, a documentation site, and an archived Zenodo DOI will be published at acceptance; the library is installable from source in the interim. The exact library version used for every credential in this paper is `pwm_dose_equivalence 0.2.2` (credential

schema `pwm-signal-equivalence/v0.2`); it will be pinned to the journal-acceptance release at submission. All scripts, configuration files, tutorial notebooks, and Docker images required to reproduce each table and figure are at https://github.com/integritynoble/low_dose_CT/tree/main/WS-2_framework. The CI pipeline that runs every reproducibility test on every commit is at https://github.com/integritynoble/low_dose_CT/actions.

Reporting Summary

A Nature Methods Reporting Summary — using the format current at submission, covering subgroup-stratified performance reporting per (U.S. Food and Drug Administration, 2024), code-and-data-availability transparency, statistical disclosure, and ethics — accompanies this manuscript as Supplementary Information. The summary cross-references the relevant Methods subsections so reviewers can locate each item in the main text.

Author contributions

[TODO: Fill at submission per CRediT taxonomy: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing-Original Draft, Writing-Review & Editing, Visualization, Supervision, Project administration, Funding acquisition.]

Competing interests

One or more authors are affiliated with the PWM Protocol Foundation, which maintains the open-source library. The library and credential format are openly licensed (Apache 2.0) and confer no privileged access; the Foundation receives no revenue from credential issuance, and no author holds a financial interest in any reconstruction method evaluated under the framework. **[TODO: At submission, each author declares any financial interest in the PWM Protocol Foundation (equity, consulting, or advisory role) or states its absence.]** The authors declare no other competing interests.

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S1 Bernstein finite-sample correction to the sample-size formula

The leading-order sample-size bound (Equation (3) in the main text) is an asymptotic-normal approximation. For credentials issued at small n (say $n < 100$) the asymptotic regime is shaky and a finite-sample bound is appropriate. This section derives the Bernstein-type bound the library reports alongside (S1).

S1.1 Setup

Let $\Delta_1, \dots, \Delta_n$ be i.i.d. random variables with $\mathbb{E}[\Delta_k] = \mu_\Delta$, $\text{Var}(\Delta_k) = \sigma_\Delta^2$, and $|\Delta_k - \mu_\Delta| \leq M$ almost surely. For task metrics bounded in $[0, 1]$ (AUC, Dice, normalised contrast-recovery), the per-patient difference $\Delta_k = a_k - b_k$ satisfies $|\Delta_k| \leq 1$, so $M = 1$ is a defensible default; for unbounded metrics (e.g. MAE on Hounsfield-unit images) M must be specified by the user.

S1.2 Bernstein's inequality

Bernstein's inequality (see, e.g., Boucheron, Lugosi & Massart 2013, Theorem 2.10) states that for $t > 0$,

$$\Pr\left[|\bar{\Delta}_n - \mu_\Delta| > t\right] \leq 2 \exp\left(-\frac{nt^2}{2\sigma_\Delta^2 + \frac{2Mt}{3}}\right). \quad (5)$$

S1.3 Substitution at $t = \varepsilon$

We require the right-hand side of (5) to be at most α . Setting $t = \varepsilon$ and rearranging,

$$n_{\text{Bernstein}} \geq \log\left(\frac{2}{\alpha}\right) \cdot \frac{2\sigma_\Delta^2 + \frac{2M\varepsilon}{3}}{\varepsilon^2}. \quad (6)$$

This is (S4) from the proofs document. The structure has two interpretable terms:

- *Variance term* $2\sigma_\Delta^2$: dominates when the per-patient differences are noisy relative to the margin. At the canonical $\alpha = 0.05$, $\log(2/\alpha) \approx 3.689$, so the variance-only Bernstein requirement is $\sim 3.689 \cdot 2\sigma_\Delta^2/\varepsilon^2 = 7.378\sigma_\Delta^2/\varepsilon^2$, which exceeds the CLT bound $z_{0.975}^2\sigma_\Delta^2/\varepsilon^2 \approx 3.842\sigma_\Delta^2/\varepsilon^2$ by a factor of ≈ 1.92 . This factor-of-two is the price Bernstein pays for being a finite-sample guarantee, not an asymptotic one.
- *Bound term* $\frac{2M\varepsilon}{3}$: dominates when the per-patient differences are nearly deterministic relative to the margin. At $\varepsilon = 0.05$ and $M = 1$, the term equals 0.0333, which is comparable to $2\sigma_\Delta^2$ at $\sigma_\Delta \approx 0.13$.

S1.4 Numerical comparison to (S1)

At the canonical AUC operating point ($\varepsilon = 0.05, \alpha = 0.05, M = 1, \sigma_\Delta = 0.10$):

Formula	$n_{\min}$
(S1) Asymptotic CLT (Eq. 3)	≥ 16
(S4) Bernstein finite-sample (Eq. 6)	≥ 50

At ($\varepsilon = 0.02, \sigma_\Delta = 0.10$) the ratio widens further: (S1) gives $n \geq 97$ vs (S4) $n \geq 308$. The widening is consistent: as ε shrinks, Bernstein's bound term contribution shrinks linearly while the

814 variance term contribution remains fixed, and the leading prefactor of $\log(2/\alpha)/z_{1-\alpha/2}^2 \approx 0.96$ on
815 the variance term combined with the factor of 2 on σ_Δ^2 gives the asymptotic ratio of ≈ 1.92 .

816 S1.5 Library behaviour and practical guidance

817 The `pwm_dose_equivalence` library computes both (S1) and (S4) when a σ_Δ hint is supplied and
818 reports both in the credential’s `sample_size_check` field. The library emits a Bernstein-explicit
819 note when the observed n falls below (S4) but above (S1) — this is the regime where the credential
820 is asymptotically valid but does not have a finite-sample guarantee, and a reviewer may want to
821 know which discipline the credential was issued under.

822 We recommend reporting the (S1) bound as the operational n -target for credentials at $n \geq 100$,
823 and (S4) as the operational target for credentials at $n < 100$. The framework remains parametric in
824 the user’s choice: the canonical defaults are suggestions, not constraints.

825 S1.6 Caveats

- 826 • Bernstein’s inequality requires $|\Delta_k - \mu_\Delta| \leq M$ almost surely. For unbounded metrics this
827 is not satisfied; in that case Hoeffding’s inequality with sub-Gaussian or sub-exponential
828 moment conditions applies, with the same general structure but a different bound term. Users
829 specifying unbounded metrics should pass $M = \infty$ to the library, which then reports only the
830 (S1) CLT bound.
- 831 • For paired AUC, $\bar{\Delta}_n$ is a U-statistic rather than a sample mean. Bernstein’s inequality for
832 U-statistics (see Hoeffding 1948; Arcones & Giné 1993) gives a structurally similar bound with
833 M replaced by the maximum placement-difference; the factor of 4 in (S3) carries through.
834 The (S4) bound as stated is therefore loose for AUC; users with $n < 100$ and AUC tasks
835 should use the DeLong closed-form CI (which has a finite-sample variance estimator) rather
836 than rely on (S4).